

Chapter 12

➤ Thermoelectric Sensors

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Summary and Consideration





Basic knowledge

- ◆ What is the definition of temperature?
- ◆ Types and working principles of thermoelectric sensors

◆ **Temperature** is a physical quantity representing the degree of coldness or hotness of an object or the environment, and also a reflection of the intensity of internal molecules' irregular thermal motion (分子无规则热运动) of an object.

◆ **The higher** the temperature, **the greater** the average **kinetic energy** (动能) of the internal **molecules** (分子).

- ◆ There are two basic types of temperature sensing: **the contact and the non-contact**.
- ◆ **Contact temperature measurement** is based on the principle of **heat balance**, which requires the sensors to be direct physical contact with the **measurand** (被测量). This method can be used to sense the temperature of solids, liquids and gases over an extremely wide temperature range.

◆ **Non-contact temperature measurement** is based on the principle of **heat radiation**, which interprets the radiant energy of a heat source in the form of energy emitted in the **infrared** (红外线的) portion of the electromagnetic spectrum.

◆ It can be used to monitor non-reflective **solids and liquids** but is not effective with **gases** because of their natural **transparency**(透明度).



Temperature scale (温标)

- ◆ Elements of the temperature scale
- ◆ Four commonly used temperature scales

- ◆ **Temperature scale** is a numerical representation (数值表述) of temperature. In order to ensure **the uniformity and accuracy** of temperature values, it is necessary to establish a standard scale (标准尺度) for measuring temperature, namely temperature scale, which specifies the reference point (fixed point) (基准点) and the units of temperature.
- ◆ **Three elements: The thermometer, the fixed points and the interpolation equation** (插值方程).

Four commonly used temperature scales

- ① Celsius temperature scale
- ② Fahrenheit temperature scale
- ③ Thermodynamic temperature scale
- ④ International Practical Temperature Scale (IPTS)

Four commonly used temperature scales

① Celsius temperature scale. Symbol “°C”

Glass mercury thermometer, freezing (0° Celsius) and boiling (100° Celsius) point of water (under the standard atmospheric pressure), linear interpolation.

Four commonly used temperature scales

② Fahrenheit temperature scale. Symbol “ F ”

Glass mercury thermometer, the mixture of ammonium chloride (氯化氨) and ice is 0°F , the freezing point and boiling point of water are 32°F and 212°F , respectively. The range is divided into 180 equal parts, each part is 1° Fahrenheit and is denoted as 1°F . linear interpolation

Four commonly used temperature scales

Note: Fahrenheit and Celsius temperature scales are all based on the relationship between the volume expansion of some substances (such as mercury) and temperature. Temperature scales are determined by empirical formulas obtained by the experimental methods, which are called empirical temperature scales with certain limitations.

Four commonly used temperature scales

③ Thermodynamic temperature scale, Symbol “K”

Thermodynamic temperature scale is the most scientific temperature scale put forward by physicist Kelvin according to the laws of thermodynamics(热力学定律).

absolute zero degree (molecules stop moving), the triple point of water (gas, liquid and solid coexist) is 273.16.

Linear interpolation.

Four commonly used temperature scales

④ International Practical Temperature Scale, IPTS

The IPTS is an international agreement temperature scale established by the International Metrology Commission, which is very similar to the thermodynamic temperature scale. temperature expressed by **interpolation function**, **high reproducible accuracy**, the **standard thermometer** prescribed is convenient to use and easy to manufacture.



Thermocouple

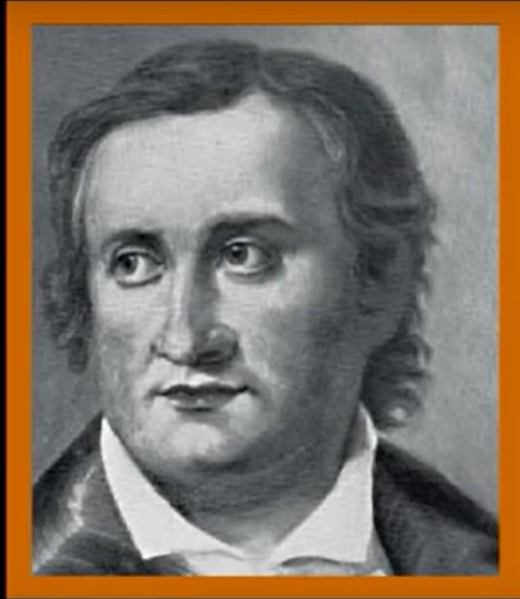
- ◆ Measuring principle of thermocouple
- ◆ Basic laws of thermocouple
- ◆ Cold junction compensation

12.4 Thermocouple

Thermocouples（热电偶） are the most widely used contact temperature sensors in the industry. Especially for **high temperature** measurement.

Advantages: simple structure, large measurement range (even $-270\sim 2800^{\circ}\text{C}$), high measurement accuracy, small thermal inertia（热惯性）, good dynamic performance, easy to use, convenient for long-distance transmission, centralized detection（集中检测） and automatic control（自动控制）.

What is Thermoelectric Effect or Seebeck Effect?

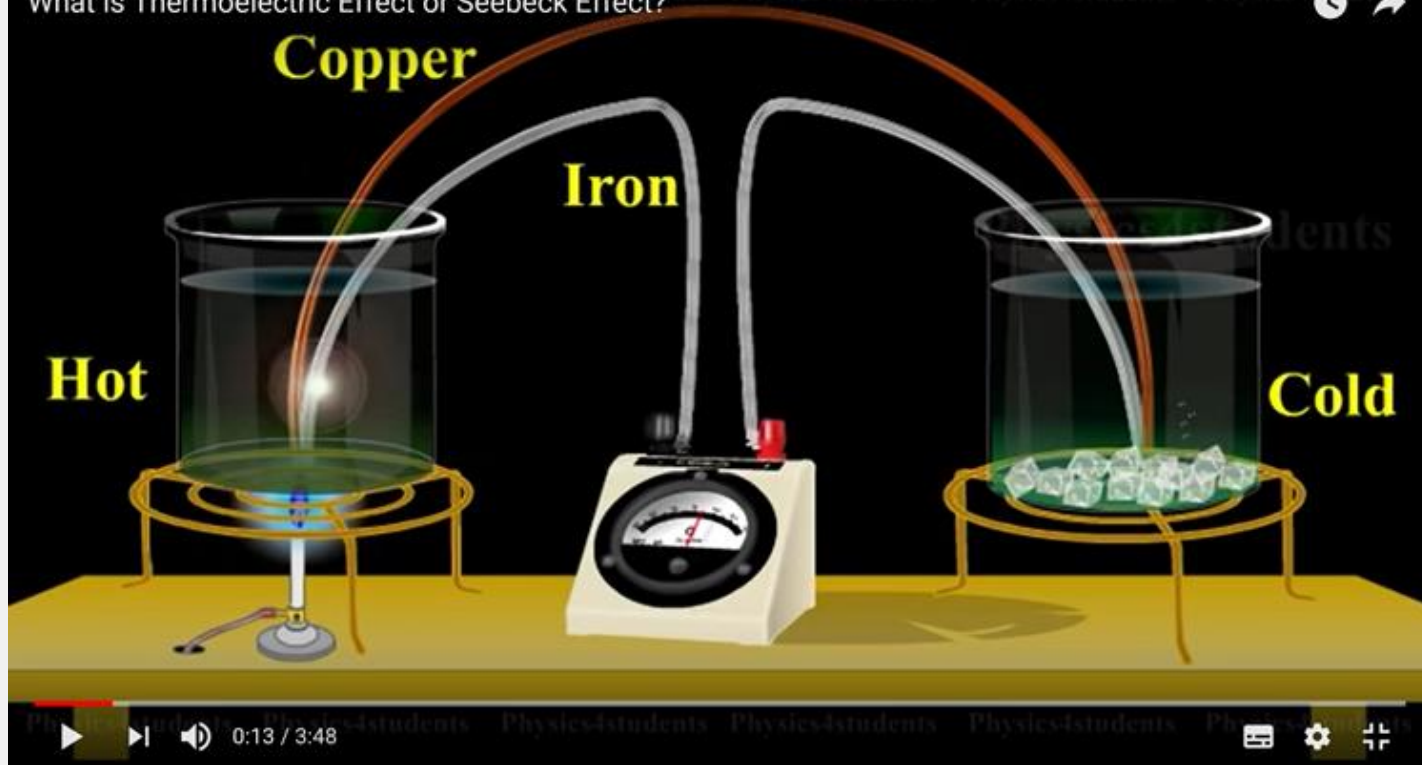


(1821)
**Thomas
Johann
Seebeck**

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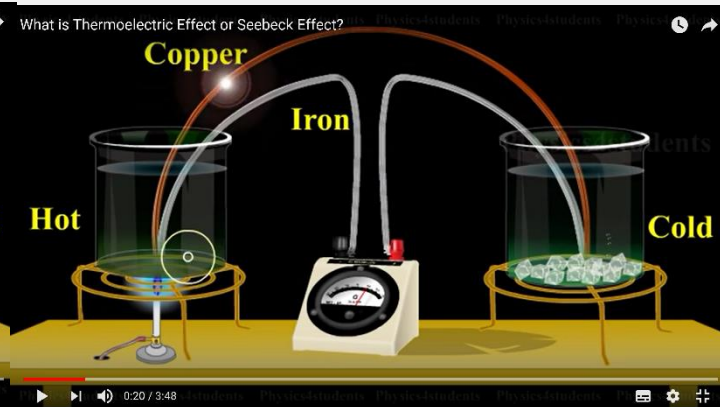
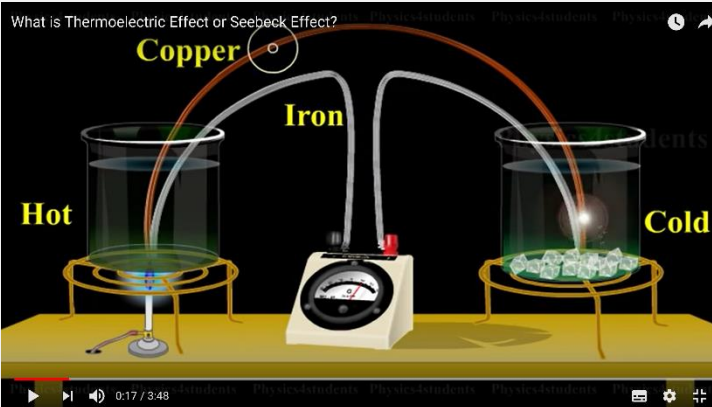
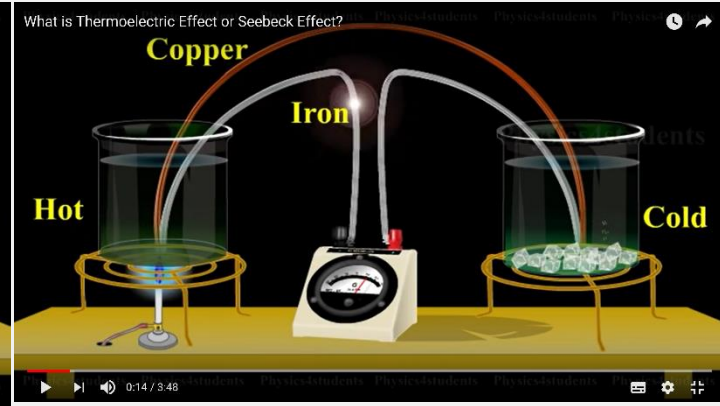
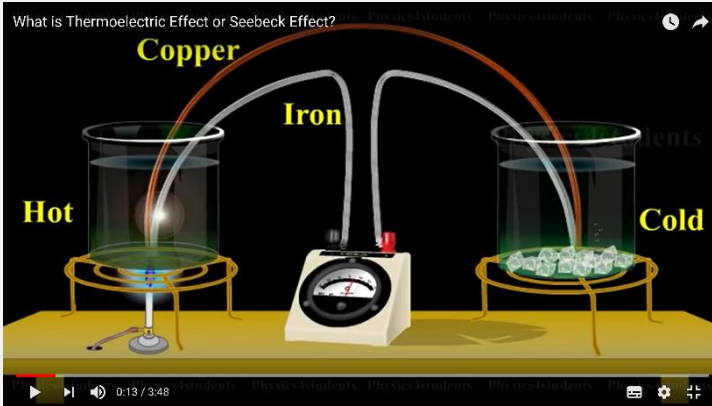


What is Thermoelectric Effect or Seebeck Effect?



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Thermoelectric Sensors

12.4.1 Measuring Principle of Thermocouple

Thermocouple is based on thermoelectric effect (Seebeck effect 塞贝克). As shown in Fig 12.9. **two different metal conductors A and B**, called thermoelectrode(热电极), **form a closed loop**; **Different junctions' temperature**, hot junction(热端) and cold junction(冷端).

The current will be generated in the loop. The corresponding electric potential is called thermoelectric potential(热电势) which is composed of the contact potential(接触电势) and the temperature difference potential(温差电势).

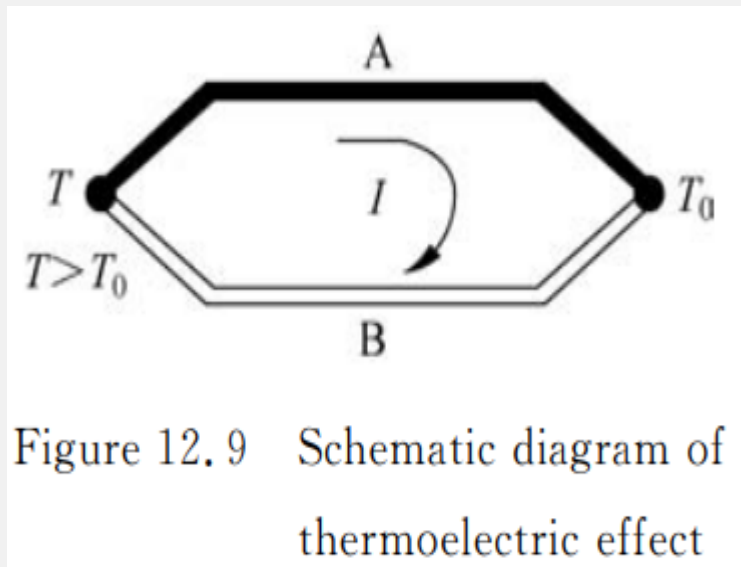


Figure 12.9 Schematic diagram of thermoelectric effect

12.4.1 Measuring Principle of Thermocouple

contact potential: the potential that is formed at the contact points by two conductors with different free electron densities is called the contact potential, also known as the Peltier potential. **Diffusion effect.**

$$E_{AB}(T) = \frac{kT}{e} \ln\left(\frac{n_A}{n_B}\right), \quad (12.6)$$

where, T is temperature of the joint point, k is the Boltzmann constant ($k = 1.38 \times 10^{-23} \text{ J/K}$), e is the electronic charge ($e = 1.6 \times 10^{-19} \text{ J/K}$), n_A and n_B are the free-charge densities of the materials A and B.

12.4.1 Measuring Principle of Thermocouple

$$E_{AB}(T) = \frac{kT}{e} \ln\left(\frac{n_A}{n_B}\right), \quad (12.6)$$

the contact potential difference depends on temperature T of the joint point and on the concentration of free electrons in metals (or depending on the property of the conductor material), but has nothing to do with the shape and size of the conductors. The contact potential is about $10^{-3} \sim 10^{-2} V$.

12.4.1 Measuring Principle of Thermocouple

temperature difference potential: is caused by the temperature difference of the same electrode's both ends, also known as the Thomson potential. **Diffusion effect.**

$$E_A(T, T_0) = \int_{T_0}^T \sigma_A dT, \quad (12.7)$$

where, σ_A is the Thomson coefficient, representing the temperature difference potential generated when the temperature difference between two ends of the same conductor is 1°C , and its value is related to the conductor material properties and temperature.

It can be seen from Eq. (12.7) that the temperature difference potential is generally much smaller than the contact potential, and its value is about 10^{-5} V.

12.4.1 Measuring Principle of Thermocouple

Thermoelectric Potential of the Thermocouple Circuit

Assuming $T > T_0$, $n_A > n_B$, the total thermoelectric potential is the **algebraic sum of the contact potentials** at the two junctions and the **temperature difference potentials** of the two conductors, as shown in Fig 12.10.

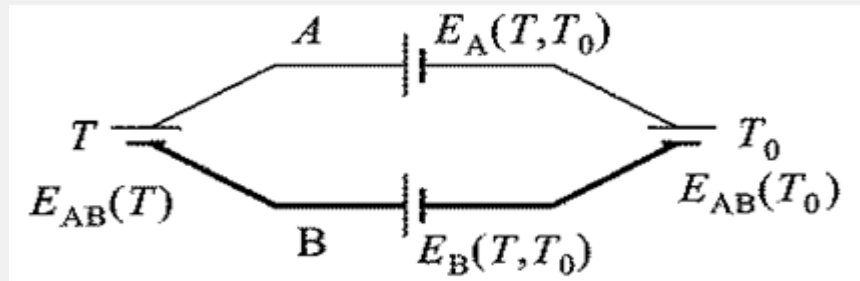


Figure 12.10 Thermoelectric Potential of Thermocouple Circuit

12.4.1 Measuring Principle of Thermocouple

Thermoelectric Potential of the Thermocouple Circuit

The total thermoelectromotive force $E_{AB}(T, T_0)$ of the thermocouple loop can be expressed as

$$\begin{aligned} E_{AB}(T, T_0) &= [E_{AB}(T) - E_{AB}(T_0)] - [E_A(T, T_0) - E_B(T, T_0)] \\ &= \frac{k}{e}(T - T_0) \ln\left(\frac{n_A}{n_B}\right) - \int_{T_0}^T (\sigma_A - \sigma_B) dT, \end{aligned} \quad (12.8)$$

12.4.1 Measuring Principle of Thermocouple

$$\begin{aligned} E_{AB}(T, T_0) &= [E_{AB}(T) - E_{AB}(T_0)] - [E_A(T, T_0) - E_B(T, T_0)] \\ &= \frac{k}{e}(T - T_0) \ln\left(\frac{n_A}{n_B}\right) - \int_{T_0}^T (\sigma_A - \sigma_B) dT, \end{aligned} \quad (12.8)$$

(1) If the two thermal electrodes of the thermocouple have the same material, even if the temperatures at the two junctions are different, the total thermoelectric potential is still zero.

12.4.1 Measuring Principle of Thermocouple

$$\begin{aligned} E_{AB}(T, T_0) &= [E_{AB}(T) - E_{AB}(T_0)] - [E_A(T, T_0) - E_B(T, T_0)] \\ &= \frac{k}{e}(T - T_0) \ln\left(\frac{n_A}{n_B}\right) - \int_{T_0}^T (\sigma_A - \sigma_B) dT, \end{aligned} \quad (12.8)$$

(2) If the temperatures of the two junctions of the thermocouple are the same, even if the materials of the two thermal electrodes A and B are different, the total thermoelectric potential in the circuit is still zero.

12.4.1 Measuring Principle of Thermocouple

$$\begin{aligned} E_{AB}(T, T_0) &= [E_{AB}(T) - E_{AB}(T_0)] - [E_A(T, T_0) - E_B(T, T_0)] \\ &= \frac{k}{e}(T - T_0) \ln\left(\frac{n_A}{n_B}\right) - \int_{T_0}^T (\sigma_A - \sigma_B) dT, \end{aligned} \quad (12.8)$$

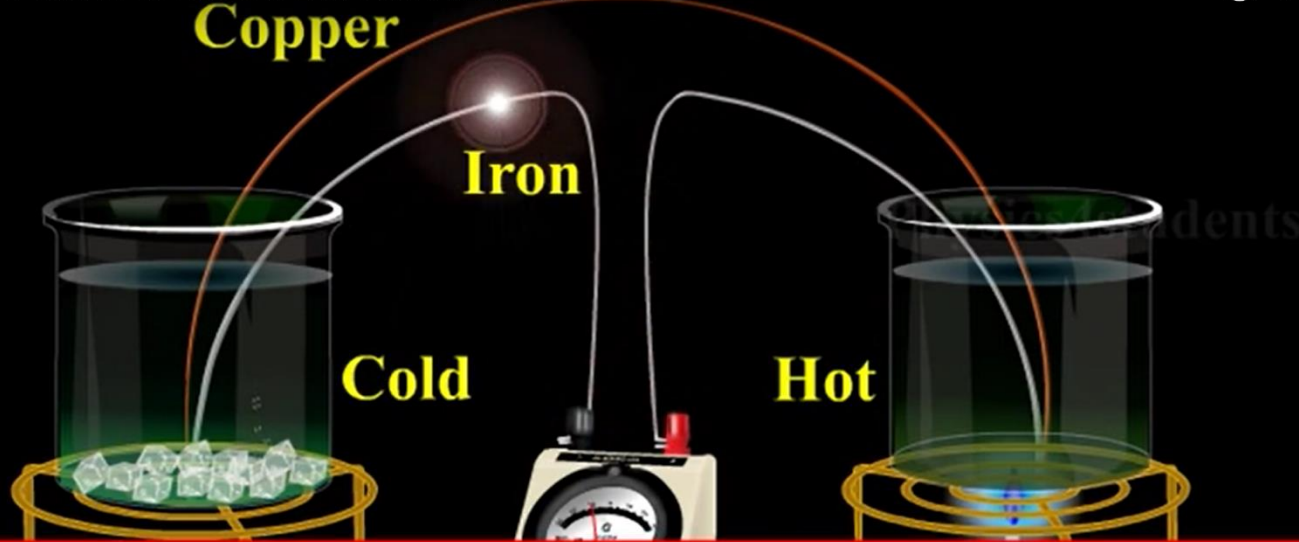
(3) The thermoelectric potential is only related to the properties of the thermal electrode material and the temperatures of the two contacts, and has nothing to do with the size and shape of the thermal electrode.

12.4.1 Measuring Principle of Thermocouple

$$\begin{aligned} E_{AB}(T, T_0) &= [E_{AB}(T) - E_{AB}(T_0)] - [E_A(T, T_0) - E_B(T, T_0)] \\ &= \frac{k}{e}(T - T_0) \ln\left(\frac{n_A}{n_B}\right) - \int_{T_0}^T (\sigma_A - \sigma_B) dT, \end{aligned} \quad (12.8)$$

(4) The electrical polarity of a thermal electrode. In the symbol ' $E_{AB}(T, T_0)$ ', it is stated that ' A ' and ' T ' represent the positive and high temperature, respectively, and ' B ' and ' T_0 ' represent the negative and low temperature, respectively. If the position of the thermal electrode is switched, the polarity of the thermoelectric potential is reversed.

What is Thermoelectric Effect or Seebeck Effect?



Hence Seebeck effect is reversible.

12.4.1 Measuring Principle of Thermocouple

Attention: there are many free electrons in metal conductors, and temperature changes have little effect on the free electron density, so the temperature difference potential is extremely small and can be ignored. Thus, the total thermoelectric potential of the thermocouple circuit can generally be approximately expressed as

$$E_{AB}(T, T_0) = E_{AB}(T) - E_{AB}(T_0), \quad (12.9)$$

$$E_{AB}(T, T_0) = f(T) - f(T_0), \quad (12.10)$$

12.4.1 Measuring Principle of Thermocouple

For ease of use, the cold junction temperature T_0 is maintained at 0°C , and the function relationship between the thermoelectric potential and the hot junction temperature can be established through experiments. The experimental data is listed in a table called the **thermocouple reference table**(热电偶分度表). (K type, 6.138mV, 150°C).

Table 12.2 Thermocouple reference table of nickel-chromium alloy and nickel-silicon alloy (K type)

$t/^\circ\text{C}$	-200	-100	0	100	200	300	400	500
E/mV	-5.891	-3.554	0	4.096	8.138	12.209	16.397	20.644
$t/^\circ\text{C}$	600	700	800	900	1,000	1,100	1,200	1,300
E/mV	24.905	29.129	33.275	37.326	41.276	45.119	48.838	52.410

Note: The cold junction is 0°C .

12.4.1 Measuring Principle of Thermocouple

The basic laws of the thermocouple

Question?

The basic laws of thermocouple provide a theoretical basis for solving engineering application problems.

① The Law of the Third Metal

The third metal conductor C is connected into the thermocouple loop composed of conductors A and B. The thermoelectric potential will not change after the inclusion of the third metal into the circuit, only if **both contact points of the third metal have equal temperature.**

(中间导体定律)

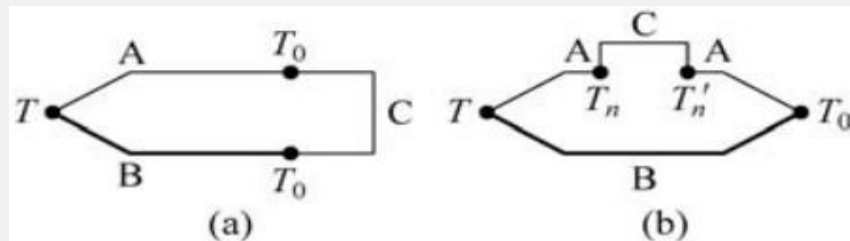


Figure 12.11 Diagram of third metal law

Thermoelectric Sensors

12.4.1 Measuring Principle of Thermocouple

② The Law of Intermediate Temperature(中间温度定律)

$E_{AB}(T, T_0)$ of the thermocouple composed of thermoelectrodes A and B at the junction temperatures (T, T_0) is equal to the **algebraic sum** of the thermoelectric potential of the thermocouple at the junction temperatures (T, T_n) and (T_n, T_0) .

It's the theoretical basis for the formulation of **thermocouple reference table** which is made when the cold junction temperature is 0°C . In addition, the law of intermediate temperature also provides a basis for the use of **compensation wires** in thermocouples.

$$E_{AB}(T, T_0) = E_{AB}(T, T_n) + E_{AB}(T_n, T_0)$$

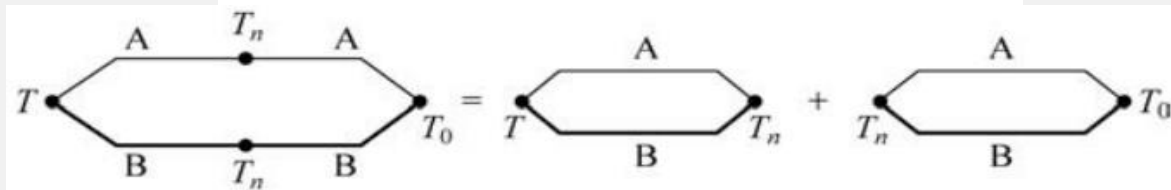


Figure 12.12 Schematic diagram of intermediate temperature law

12.4.1 Measuring Principle of Thermocouple

Example 12.1

A thermocouple composing of nickel-chromium alloy and nickel-silicon alloy is employed to measure the furnace temperature. Its cold end temperature is $T_n = 20^\circ\text{C}$, the thermoelectric potential measured by the meter is $E(T, 20^\circ\text{C}) = 7.33 \text{ mV}$. What is the furnace temperature?

Solution: checking the reference table of nickel-chromium alloy and nickel-silicon alloy thermocouple,

$$E(T_n, T_0) = E(20^\circ\text{C}, 0^\circ\text{C}) = 0.798 \text{ mV}.$$

According to the Formula (12.11),

$$E(T, T_0) = E(T, T_n) + E(T_n, T_0) = 7.33 \text{ mV} + 0.798 \text{ mV} = 8.128 \text{ mV}.$$

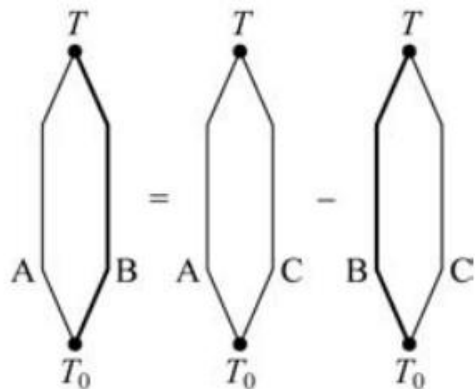
Checking the reference table again, the furnace temperature is $T = 200^\circ\text{C}$.

Thermoelectric Sensors

12.4.1 Measuring Principle of Thermocouple

③ Standard Electrode Law (标准电极定律)

If the thermoelectric potential of thermocouple composed of conductor A and conductor C and the potential of the thermocouple composed of conductor B and conductor C are known respectively, the thermoelectric potential of the thermocouple composed of conductor A and conductor B is the **algebraic sum** of the thermoelectric potentials of the two thermocouples.



$$E_{AB}(T, T_0) = E_{AC}(T, T_0) + E_{CB}(T, T_0) = E_{AC}(T, T_0) - E_{BC}(T, T_0)$$

The conductor C is called the **standard electrode**, conductor C is usually made of **pure platinum wire**. The standard electrode law greatly simplifies the selection of thermoelectrodes.

12.4.2 Type, Structure and Measuring Circuit of Thermocouple

Type of Thermocouple

At present, there are more than 300 kinds of materials suitable for making thermoelectrodes, among which 40-50 kinds are widely used. Commonly used thermoelectrode materials are divided into three categories: **precious metals**, **ordinary metals** and **non-metallic materials**. According to the International Temperature Scale (ITS-90), there are 8 types of industrial standardized thermocouples.

Table 12.3 Parameters of some industrial thermocouples

Type	Metal (+)	Metal (—)	Sensitivity $/\mu\text{V} \cdot ^\circ\text{C}^{-1}$	Temperature range/ $^\circ\text{C}$	Tolerance ΔT at 750°C	Properties
K	Chromel Ni-Cr	Nickel	40.5	-270 to $+1,370$	class 1: 3.0°C class 2: 5.6°C	Good linearity
S	Platinum	Pt-Rh10 90%Pt + 10%Rh	6.4	-50 to $+1,760$	class 1: 1°C class 2: 1.9°C	Good accuracy
J	Iron	Copper-nickel	51.7	-210 to $+1,200$	class 1: 3°C class 2: 5.6°C	Cheap, good sensitivity
T	Copper	Copper-nickel	49	-200 to $+400$	class 1: 1.4°C class 2: 2.6°C	Cheap, good sensitivity

Note: The tolerance for the T type thermocouple is standardized only in the temperature range from -40°C to $+350^\circ\text{C}$; the value of ΔT for it is given for $T = +350^\circ\text{C}$.

12.4.2 Type, Structure and Measuring Circuit of Thermocouple

Structure of Thermocouple

The structure of the thermocouple differs depending on the purpose. The structure of the common thermocouple includes **ordinary thermocouple**, **armored thermocouple**, and **thin film thermocouple**. (普通型热电偶, 铠装热电偶, 薄膜热电偶) The ordinary thermocouple is composed of **thermal electrode**, a **protective sheath** (绝缘套管), a **protective covering** (保护套管) and a **terminal box** (接线盒), etc. Used for gas, steam and liquid and other medium.

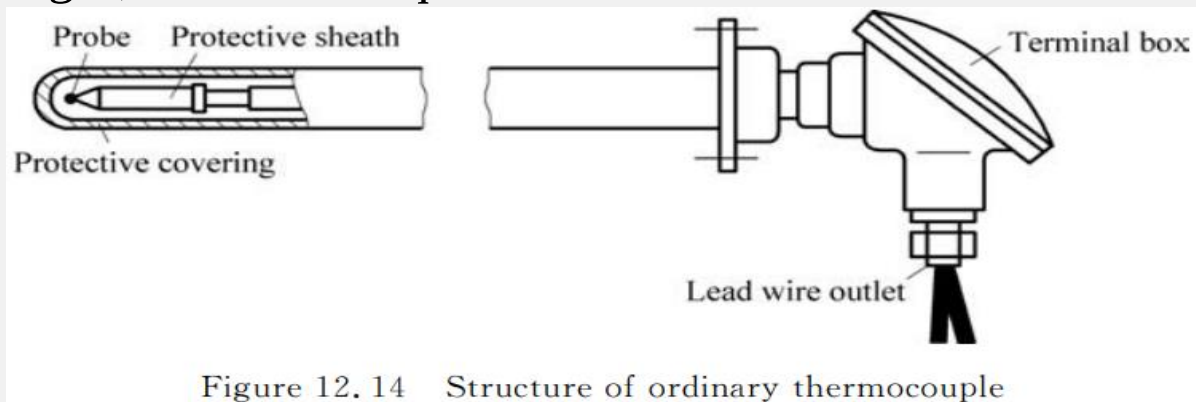


Figure 12.14 Structure of ordinary thermocouple

12.4.3 Cold Junction Compensation

Generally, the cold junction temperature is need to be kept at a **consistent**. In practical applications, the cold junction is often at room temperature or very close to the heat source, which is susceptible to the influence of the measurement object and environmental temperature fluctuations, **making it difficult to keep the cold junction temperature constant**.

① Cold junction thermostatic method

The constant temperature method is to place the cold junction of the thermocouple in a device with a constant temperature.

$$E_{AB}(T, T_0) = E_{AB}(T, T_n) + E_{AB}(T_n, T_0).$$

Thermoelectric Sensors

12.4.3 Cold Junction Compensation

② Compensation wire method

The compensating wire is often used to lengthen the thermocouple to extend the cold junction far away from the heat source.

The compensating wire **large diameter and high conductivity** to reduce the influence of its own resistance.

Key points:

The compensating wire can only be **matched with** the corresponding thermocouple;

The positive and negative **polarity** cannot be wrongly connected;

The two junctions of thermocouple and compensation wires should have **the same**

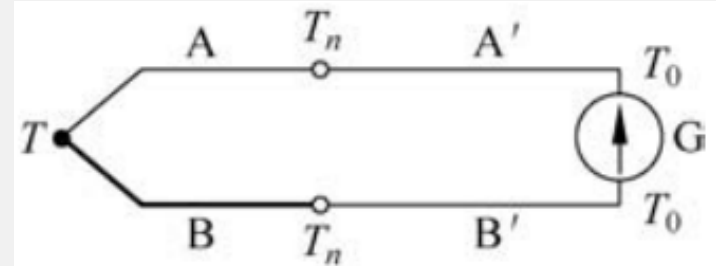


Figure 12.17 Compensation wire method

12.4.3 Cold Junction Compensation

③ Compensation bridge method

The compensation bridge method uses the potential (or compensation voltage) generated by the unbalanced bridge caused by temperature change to automatically compensate the thermoelectric potential change due to the temperature fluctuation of the cold junction, and the device is called the **cold junction temperature compensator**.

The resistor R_t is sensitive to temperature. Others are not. The bridge is powered by a DC stabilized power supply.

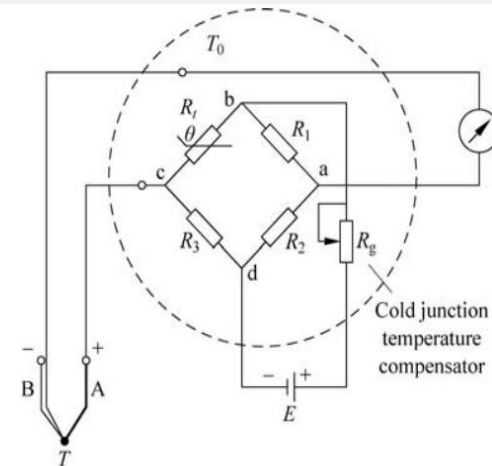


Figure 12.18 The working principle diagram of compensation bridge method

Thermoelectric Sensors

12.4.4 Calibration of Thermocouple

Considering that the thermocouple is used for a period of time, due to the influence of physical factor, chemical factor and other factors, the thermoelectric characteristics of the thermocouple will change, and the measurement accuracy will decrease. Therefore, **regular calibration is required.**

There are two methods usually employed to calibrate the thermocouple. The **fixed-point method(定点法)** and the **comparison method(比较法)**.

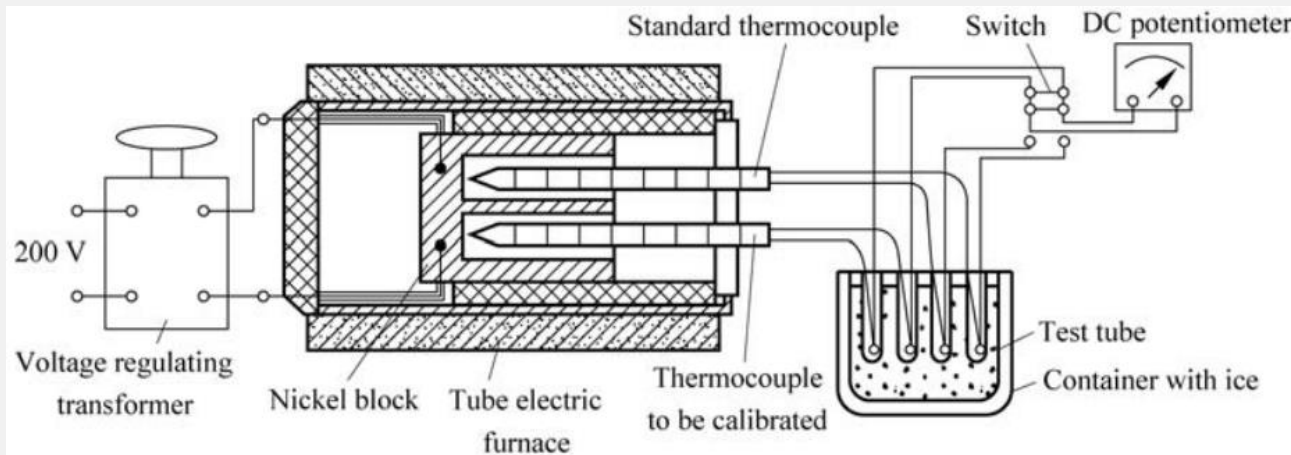


Figure 12.19 The schematic diagram of thermocouple calibration

3、常用热电偶及热电势的测量

➤常用热电偶

- 常用热电偶可分为标准热电偶和非标准热电偶两大类。
- **标准热电偶**是指国家标准规定了其热电势与温度的关系、允许误差、并有统一的标准分度表的热电偶，它有与其配套的显示仪表可供选用。
- **非标准热电偶**在使用范围或数量级上均不及标准化热电偶，一般也没有统一的分度表，主要用于某些特殊场合的测量。

适于制作**热电偶的材料**有300多种，其中广泛应用的有40～50种。常用8种标准化热电偶为：

8种标准化热电偶		
型号标志	材料	使用温度℃
S	铂铑10——铂	-50～1768
R	铂铑13——铂	-50～1768
B	铂铑30——铂铑6	0～1820
K	镍铬——镍硅	-270～1372
N	镍铬硅——镍硅	-270～1300
E	镍铬——铜镍合金（康铜）	-270～1000
J	铁——铜镍合金（康铜）	-210～1200
T	铜——铜镍合金（康铜）	-270～400

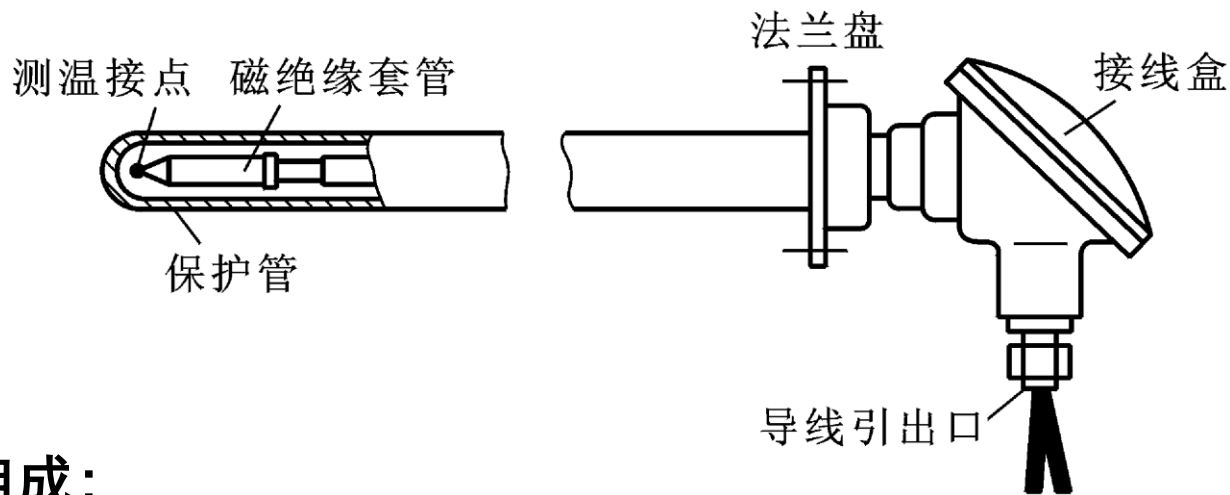
➤ **铂铑10—铂热电偶**：性能稳定，准确度高，可用于基准和标准热电偶。热电势较低，价格昂贵，不能用于金属蒸汽和还原性气体中；

➤ **铂铑30—铂铑6热电偶**：较铂铑10—铂热电偶更具较高的稳定性和机械强度，最高测量温度可达1800℃，室温下热电势较低，可作标准热电偶，一般情况下，不需要进行补偿和修正处理。由于其热电势较低，需要采用高灵敏度和高精度的仪表；

➤ **镍铬—镍硅或镍铬—镍铝热电偶**：热电势较高，热电特性具有较好线性，良好的化学稳定性，具有较强的抗氧化性和抗腐蚀性。稳定性稍差，测量精度不高。

➤ **镍铬—考铜热电偶**：热电势较高，电阻率小，适于还原性和中性气氛下测量，价格便宜，测量上限较低；

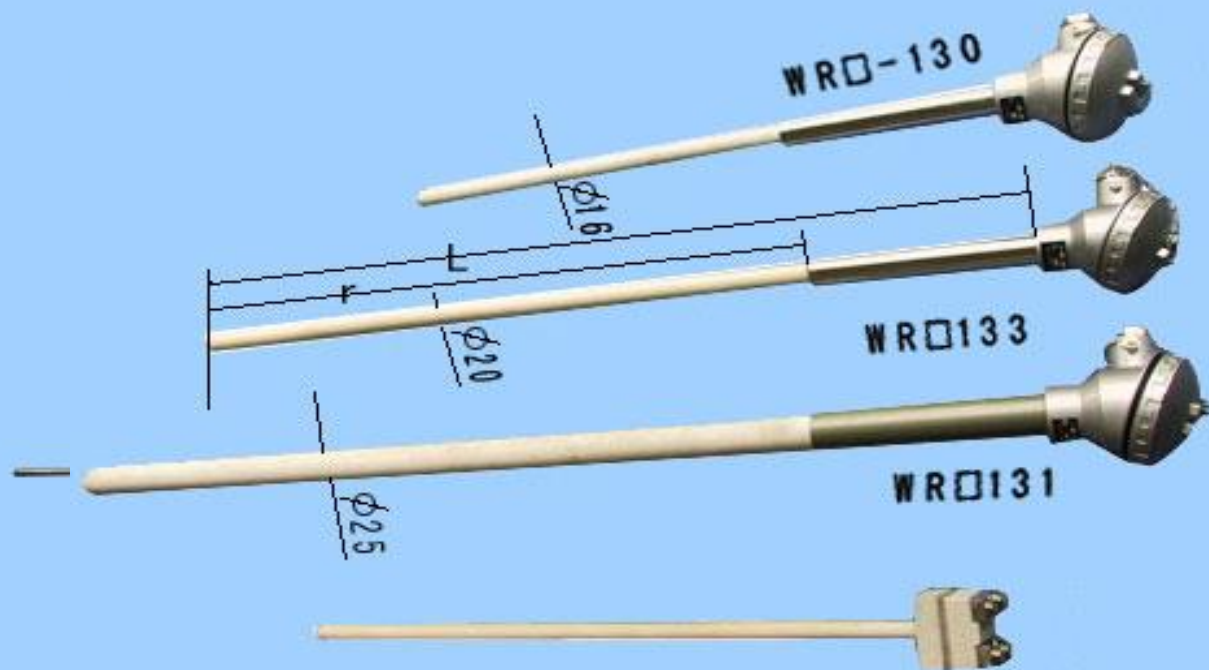
➤ **镍铬—康铜热电偶**：热电势较高，价格低。高温下易氧化，适于低温和超低温测量。



组成：

- 热电极（材料的热电特性稳定）；
- 绝缘套管（防止电极间短路）；
- 保护套管（隔离电极与介质，免受机械损伤或化学腐蚀等）；
- 接线盒。

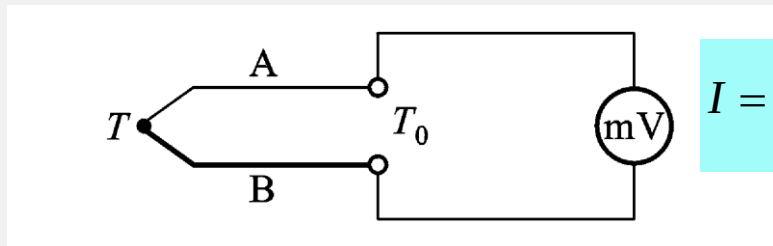
产品



➤ 热电势的测量

热电势在mv范围，通常用**动圈式仪表**、**电位差计**、**示波器**和**数字式测温仪表**等测量。

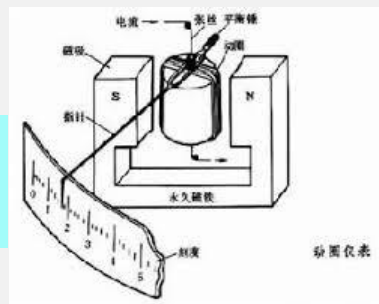
◆ 动圈式仪表实际上是一个**磁电式毫伏计**，它是利用电流流过仪表动圈时，动圈在磁场的作用下带动指针偏转的原理进行工作的。**精度不高**。



$$I = \frac{E_{AB}(T, T_0)}{R_t + R_l + R_G}$$

仪表指示电压为:

$$U = I R_G = \frac{E_{AB}(T, T_0)}{R_t + R_l + R_G} R_G$$



$$R_G \gg R_t + R_l$$

$$U \approx E_{AB}(T, T_0)$$

4、热电偶的冷端温度补偿

- ◆ 热电偶回路的热电势是两个接点温度的函数，为此，必须使冷端温度保持不变，其输出热电势才是热端温度的单值函数。
- ◆ 在实际测量中，热电偶的两端距离很近，冷端温度易受测量对象和环境温度波动的影响，而难以保持恒定，从而引入测量误差。
- ◆ 冷端温度补偿：冷端温度恒定；补偿冷端温度 $t_0 = 0^\circ\text{C}$ 。补偿方法： 0°C 恒温法；修正法；补偿导线法；补偿电桥法等。

0°C恒温法

将热电偶冷端放在冰和水混合的容器中，保持冷端为0°C不变。这种方法精度高，但在工程中应用很不方便，一般在实验室用于校正标准热电偶等高精度温度测量。

修正法

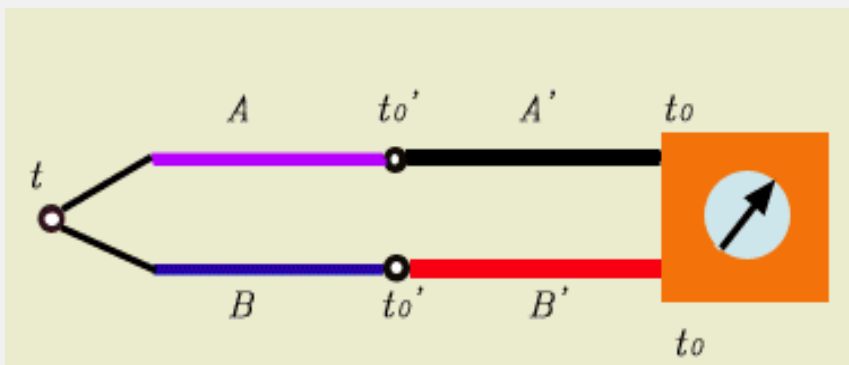
实际使用中，设法使冷端温度保持不变（放置在恒温器中），然后采用冷端温度修正的方法，可得到冷端为0°C时的热电势。

根据**中间温度定律**， $E_{AB}(T, T_0) = E_{AB}(T, T_n) + E_{AB}(T_n, T_0)$ ，因为保持温度 T_n 不变，因而 $E_{AB}(T_n, 0) =$ 常值，该值可以从热电偶分度表中查出。

测量的热电势与查表得到的相加，就可得到冷端为0°C时的热电势，然后再查热电偶分度表，便可得到被测温度 T 。

补偿导线法——延长热电极法

将热电偶的自由端引至显示仪表，而显示仪表放在恒温或温度波动较小的地方。



图示：补偿导线的应用

A, B —热电极； A', B' —补偿导线

t_0' —原参考端温度； t_0 —新参考端温度

采用某两种导线组成的热电偶补偿导线，在一定温度范围内（ $0\sim 100^{\circ}\text{C}$ ）具有与所连接的热电偶**相同**的热电性能。

不同的热电偶要配不同的导线，极性也不能接错。